MeanFlow — English Presentation Script

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Slide 1 — Introduction

Good morning everyone. My name is Zhefeng Lin.

My project focuses on Mean Flows for One-step Generative Modeling.

I will first explain why I selected this paper and introduce its main idea. Then, I will present my research question: can a different interval normalization improve MeanFlow training?

Slide 2 — Why Did I Choose MeanFlow?

I chose this paper for three reasons.

First, it addresses a practical problem: reducing the computational cost of image generation.

Second, its mathematical idea is clear. An identity relating average and instantaneous velocities leads directly to a neural-network training objective.

Third, it raises an interesting optimization question. The paper normalizes displacement by the time interval, but is this necessarily the best choice for training?

The images shown here are examples from the paper, not my own experimental results.

Slide 3 — Connection to Machine Learning

MeanFlow is a generative model that learns to transform noise into images.

Flow Matching learns instantaneous velocity and typically requires multiple evaluations to integrate a trajectory. MeanFlow instead learns average velocity over an interval, allowing generation with one evaluation.

The figure illustrates conditional and marginal flows. Importantly, straight conditional paths do not necessarily produce straight marginal trajectories.

My project connects three machine-learning topics: function representation, neural-network optimization, and generative performance.

Slide 4 — How MeanFlow Works

MeanFlow defines average velocity as the displacement along a trajectory divided by the interval length.

Differentiating this definition gives the identity shown here: instantaneous velocity equals average velocity plus the interval length times its total derivative.

The diagram distinguishes instantaneous velocity, which follows the tangent, from average velocity, which follows the displacement direction.

During training, the total derivative is computed using a Jacobian-vector product. The resulting regression target is treated with stop-gradient.

At inference, one average-velocity prediction gives the full noise-to-image displacement.

Slide 5 — Generalizing the Normalization

My proposed direction replaces division by the interval length with division by a chosen function, f.

The physical displacement remains unchanged: it is f times the predicted field.

Differentiating gives the generalized identity shown on this slide.

I will consider smooth functions with suitable boundary behavior and positive derivatives. This avoids singular regression targets and preserves a finite local limit.

Sampling still requires only one network evaluation.

Slide 6 — Research Hypothesis

Although different normalizations can represent the same physical flow, they need not produce identical training updates.

Because the regression target uses stop-gradient, changing f changes the target scaling and interval-dependent updates.

My hypothesis is that some choices could improve stability or sample quality.

However, an apparent improvement might come only from reweighting time intervals. Therefore, a matched-reweighting control is essential.

I am proposing an investigation, not claiming an improvement yet.

Slide 7 — Experimental Plan

I will use CIFAR-ten and the approximately fifty-five-million-parameter U-Net from the paper.

I will compare the original normalization, two alternatives, and matched-reweighting controls, while keeping the architecture and training budget consistent.

Evaluation will include FID on fifty thousand generated images, training time, gradient stability, and variation across three random seeds.

A squared-loss ablation will help isolate the reweighting effect. ImageNet is an optional extension if sufficient compute is available.

Slide 8 — Progress and Next Steps

So far, I have studied the paper, derived the generalized identity and matched-update relation, and checked these formulas numerically on analytic examples.

Neural-network experiments have not started yet.

Next, I will verify the automatic derivatives, reproduce the baseline, and run controlled comparisons.

The final goal is to determine whether normalization provides benefits beyond simple reweighting—and explain the result even if it is negative.

Thank you.